

JAEHYUNG LEE

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EDUCATION

Johns Hopkins University, Baltimore, MD
Ph.D. in Materials Science and Engineering

January 2025 – Present

Columbia University, New York, NY
M.S. in Chemical Engineering

Graduation: December 2024

Pennsylvania State University, State College, PA
B.S. in Chemical Engineering

Graduation: May 2023

WORK EXPERIENCE

EnerSys, Reading, PA
Advanced Quality Engineer Intern

Jan 2022 – Jun 2022

- Worked as an engineering intern performing failure mode effective analyses of Li-ion and lead-acid battery cells.
- Resolved a total of 204 production part deviation cases and gave monthly updates to project manager and C-suite.
- Partnered with diverse teams to generate a production line Li-ion battery control plan for a next-generation product.

United States Forces Korea, Camp Humphreys, South Korea
KATUSA (E-5 Sergeant)

Jun 2018 – Feb 2020

- Served 21 months of mandatory military service as a KATUSA (Korean Augmentation to the United States Army).
- Acted as a personal interpreter for a U.S. Army captain during joint Republic of Korea and U.S. Army operations.

RESEARCH EXPERIENCE

Johns Hopkins University, Baltimore, MD
Doctoral Student, Choudhary Lab

Jan 2025 – Present

- Developing materials discovery workflows integrating DFT data with machine learning for energy applications.
- Building and applying neural network models to predict properties such as energies, forces, and electronic behavior.
- Designed agentic AI platforms that combine LLMs with scientific APIs to enable automated design workflows.
- Managing the group's Linux HPC server and optimizing DFT and ML workloads with GPU-accelerated inference.
- Curating and maintaining materials datasets for training, benchmarking, and validating machine learning models.

Columbia University, New York, NY
Graduate Research Assistant

Jan 2024 – Dec 2024

- Pursued atomistic scale research in Urban Lab, using computational tools to model and analyze materials behavior.
- Trained ML interatomic potentials of lithium nickel oxide and titanium oxide to perform efficient calculations.
- Awarded the Société de Chimie Industrielle Scholarship to conduct research and present at the poster session.

Pennsylvania State University, State College, PA
Undergraduate Research Assistant

Aug 2020 – May 2023

- Performed research of analyzing catalytic sites in metal oxides based on DFT results in Janik and Alexopoulos Labs.
- Mentored newly joined undergraduate students on how to utilize computational methods and machine learning tools.

TEACHING EXPERIENCE

Johns Hopkins University, Baltimore, MD

Jan 2026 – Present

- Teaching Assistant, EN.510.312 Physical Chemistry of Materials I: Thermodynamics
Spring 2026
- Teaching Assistant, EN.520.677 Applied Quantum Computing
Fall 2026

EXTRACURRICULAR ACTIVITIES

Johns Hopkins Pava Center for Entrepreneurship, Baltimore, MD
Spark Accelerator, Fall 2026 Cohort

Sep 2026 – Present

- Selected to develop Actuator (actuato.com), a marketplace where AI agents hire local workers for physical tasks.
- Built and launched the platform, including the web app, API, and an MCP server that lets AI assistants post jobs.

- Represented the department as an ambassador by communicating with admitted students and sharing program insight.
- Assisted the development of the MS handbook and participated as a panelist at the department master's colloquium.

American Institute of Chemical Engineers PSU Chapter, State College, PA
Vice President

Aug 2020 – May 2023

- Hosted monthly mentorship sessions and academic discussions among chemical engineering undergraduate students.
- Accompanied professors to represent Penn State University in the 2021 and 2022 AIChE National Conferences.

Engineers Without Borders PSU Chapter, State College, PA
Development Committee Chair

Aug 2020 – May 2023

- Led a subgroup in the club to design and facilitate the construction of a water supply facility in Namutamba, Uganda.
- Managed corporate networking and organized fundraising outreach events as chair of the development committee.

PUBLICATIONS

1. J. Lee, *et al.* **AGAPI-Agents: An Open-Access Agentic AI Platform for Accelerated Materials Design on AtomGPT.org.** *J. Phys. Chem. Lett.* **17**(26), 7221–7231 (2026). DOI: 10.1021/acs.jpcclett.6c00837.
2. J. Lee, *et al.* **Lessons Learned from the 2025 Agentic AI for Science Hackathon.** *Mach. Learn.: Sci. Technol.* **7**(4), 040501 (2026). DOI: 10.1088/2632-2153/ae7f6a.
3. J. Lee, *et al.* **BatteryMat: A hierarchical machine-learning and DFT framework for average-voltage screening of lithium-ion cathode materials.** *npj Comput. Mater.*, under review. Preprint: arXiv:2607.06645.
4. J. Lee, *et al.* **ALIGNN 2.0: A Unified Line-Graph Neural Network Framework for Materials Screening, Force Fields, Inverse Design, Spectroscopy, and Microscopy.** *Nat. Mach. Intell.*, under review. Preprint: arXiv:2609.19487.
5. F. Abel, J. Lee, *et al.* **RamanGPT: Bidirectional Mapping Between Crystal Structures and Raman Spectra with Graph Neural Networks and Generative Transformers.** *Sci. Rep.*, under review. Preprint: arXiv:2606.03764.
6. A. Ajith, J. Lee, *et al.* **SlaKoNet-VQD: A universal Slater-Koster tight-binding Hamiltonian for variational quantum band-structure calculations on near-term hardware.** *Sci. Rep.*, under review. Preprint: arXiv:2607.09761.
7. C. Campbell, J. Ely, J. Lee, *et al.* **Hybrid DiffractGPT-Rietveld Refinement Framework for Automated X-ray Diffraction Analysis.** *Sci. Rep.*, under review. Preprint: arXiv:2607.08890.
8. H. Neralla, J. Lee, *et al.* **Hallucination Detector: A hybrid LLM and Semantic Scholar tool calling for detecting hallucination in scientific literature on AtomGPT.org.** *Integr. Mater. Manuf. Innov.*, under review. Preprint: arXiv:2607.09774.
9. G. Han, T. Li, X. Xu, J. Lee, *et al.* **The search for high-entropy fuel-cell catalysts using disorder descriptors.** *Nano Futures* **9**(4), 045001 (2025). DOI: 10.1088/2399-1984/ae19b0.
10. B. Slautin, K. Barakati, U. Pratiush, C. Lowe, C. Bodinger, B. Cossairt, M. Ahmadi, A. Houton, T. Bazhironov, J. Lee, *et al.* **Accelerated materials discovery through random-library exploration by automated scanning transmission electron microscopy.** *Digit. Discov.*, advance article. DOI: 10.1039/d6dd00186f.

SELECTED PRESENTATIONS

1. “SlaKoNet DB: A Cross-Domain Tight-Binding Database of Electronic Structure,” Workshop on Recent Developments in Electronic Structure (ESW), Madison, WI, June 2026.
2. “Agentic AI for Materials Design with AGAPI,” Materials Research Society (MRS) Spring Meeting & Exhibit, Honolulu, HI, April 2026.
3. “BatteryMat: A Machine Learning Accelerated DFT Framework for Screening Battery Cathode Materials,” National Institute of Standards and Technology (NIST) Quantum Matters in Materials Science (QMMS) Workshop, Gaithersburg, MD, February 2026.

4. “Analysis of Cathode Materials for Lithium-Ion Batteries Using Machine Learning Potentials,” Société de Chimie Industrielle Biennial International Palladium Medal Ceremony, New York, NY, May 2024.

HONORS AND AWARDS

Fellowship – **Amazon AI PhD Fellow**, Amazon.

September 2026 – Present

Honor Society – **Sigma Xi Full Member**, Sigma Xi.

July 2026 – Present

Scholarship – **Société de Chimie Industrielle Scholar**, Société de Chimie Industrielle.

May 2024 – August 2024